NATIONAL OPEN UNIVERSITY OF NIGERIA,
PLOT 91, CADASTRAL ZONE, UNIVERSITY VILLAGE, JABI – ABUJA
FACULTY OF SCIENCES

Course Code: CIT 383
Course Title: Introduction to Object-Oriented Programming
Time: 2 Hours
Credit: 2 Units|
Instruction: Attempt 4 questions. Question1 is Compulsory and any other three (3)
Question 1

(a). Explain Inheritance, state its usefulness and give an example of its applicable area. (5marks)
(b). Differentiate between the following:
(i) a constructor and a method (3marks)
(ii) a no argument constructor and an argument constructor (2marks)
(c). Describe two functionality of a downcasting and state its syntax (5marks)
(d). (i) what do you understand by Operator overloading (2marks)
(ii) Identify two reasons why Java does not support operator overload (3marks)
(e). Write the syntax for overloading the true and false operators
(5marks)

Question 2

(a) (i) What does it mean to say that a subclass "overrides a method"
(2marks)
(ii) Identify the various ways a subclass specializes its behaviour with respect to its superclass (3marks)
(b). Describe abstract class in Java (5marks)
(c). Write the syntax for an abstract class in Java
(5marks)

Question 3

(a). Briefly summarize any four properties of an implicit cast (4marks)
(b). Identify the three (3) key restrictions to overloading a conversion operator (6marks)
(c). Describe using appropriate accessor declaration, a one-dimensional indexer (5marks)

Question 4

Create a class Invoice that implements the Payment interface using the following guide:
(a) This class should show the product number and description, (5marks)
(b) it should also show the product quantity and the price of product. (5marks)
(c) Method getPaymentAmount should be implemented in this class to calculate the amount on the invoice. (5marks)

Question 5

(a). Identify any six properties of Interface in OOP (6marks)
(b). Outline the properties of an Abstract Data Type (ADT) (4marks)
(c). Enumerate on the features of Encapsulation attribute in OOP (5marks)